Three new problems from the Erdős corpus

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Abstract

I introduce three problems arising from a systematic study of the Erdős problem corpus. (1) *Density-relaxed Schur numbers* $\mathrm{DS}(k,\alpha)$ interpolate between Schur numbers and density Ramsey theory; we determine $\mathrm{DS}(2,\alpha)$ for all α, revealing a sharp phase transition at $\alpha = 1/2$, with a complete Lean 4 formalization. (2) *Coprime Ramsey numbers* $R_{\mathrm{cop}}(k)$ measure Ramsey forcing on the coprime graph; we prove $R_{\mathrm{cop}}(3) = 11$ by exhaustive extension. (3) I show that the set of primes, which maximizes $\sum 1/(a \log a)$ over primitive sets (Lichtman [5]), is *not* optimal for coprime pair counts, and give a lower bound $M^*(n) \geq \frac{64}{9\pi^2}\binom{\lceil n/2 \rceil}{2}(1+o(1))$.

MSC 2020: 05D10, 05C55, 11B75, 11N25.

1 Introduction

A systematic study of the 1,135 problems in the Erdős problem database [8] reveals natural questions at the intersection of Erdős's interests—Schur numbers, coprime graphs, primitive sets—that do not appear to have been formulated. I present three such problems with solutions and computational results. All code and Lean formalizations are publicly available.[1]

2 Density-relaxed Schur numbers

A *Schur triple* is a triple (a,b,c) with $a+b=c$ and $a,b \geq 1$ (so $a=b$ is permitted).

Definition 2.1. For $k \geq 1$ and $\alpha \in (0,1]$, a k-coloring of $[N]$ is (k,α)-*avoiding* if every color class is both sum-free and has at most αN elements. Define

$$\mathrm{DS}(k,\alpha) \;=\; 1 + \max\{N \geq 1 : \text{a } (k,\alpha)\text{-avoiding coloring of } [N] \text{ exists}\},$$

with $\mathrm{DS}(k,\alpha) = 1$ if no such N exists.

When $\alpha = 1$ the density constraint is vacuous, so $\mathrm{DS}(k,1) = S(k)+1$ where $S(k)$ is the k-th Schur number [7]. When $\alpha < 1/k$, pigeonhole forces some class to exceed αN elements for all $N \geq 1$, so $\mathrm{DS}(k,\alpha) = 1$. Since Schur triples force at $S(k)+1$ independently of density, we always have $\mathrm{DS}(k,\alpha) \leq S(k)+1$.

Theorem 2.2. $\mathrm{DS}(2,\alpha) = 5$ *for all* $\alpha \geq 1/2$, *and* $\mathrm{DS}(2,\alpha) = 1$ *for* $\alpha < 1/2$. *In particular* $\mathrm{DS}(2,\frac{1}{2}) = 5 = S(2)+1$.

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[1]Code and data: `doi.org/10.5281/zenodo.18638635`

Proof. For $\alpha < 1/2$: in any 2-coloring of $[N]$, some class has $\lceil N/2 \rceil > \alpha N$ elements (since $\alpha < 1/2$), so no $(2, \alpha)$-avoiding coloring exists for any N.

For $\alpha \geq 1/2$: the coloring $C_1 = \{1, 4\}$, $C_2 = \{2, 3\}$ of $[4]$ has both classes sum-free with $|C_i| = 2 \leq \alpha \cdot 4$ (using $\alpha \geq 1/2$). So $N = 4$ is $(2, \alpha)$-avoiding. Since $S(2) = 4$, every 2-coloring of $[5]$ contains a Schur triple, so $N = 5$ is not avoiding. Hence $\mathrm{DS}(2, \alpha) = 1 + 4 = 5$. □

Remark 2.3. The density-forcing threshold ($\alpha = 1/2 = 1/k$ for $k = 2$) coincides with the Schur-forcing threshold ($S(2) + 1 = 5$), collapsing $\mathrm{DS}(2, \cdot)$ to a step function. For $k \geq 3$, these thresholds may separate, yielding a non-trivial interpolation regime.

Remark 2.4. Theorem 2.2 has been fully formalized in Lean 4 (Mathlib), with 12 theorems proved and 0 `sorry` statements. The formalization is in `lean/NPG2_DensitySchur.lean`.

Open Problem 2.5. Determine $\mathrm{DS}(k, \alpha)$ for $k \geq 3$ and $\alpha \in (1/k, 1)$. Is there a non-trivial interpolation regime, i.e., does $\mathrm{DS}(k, \alpha) < S(k) + 1$ for some $\alpha > 1/k$?

3 Coprime Ramsey numbers

Definition 3.1. The *coprime graph* $G_{\mathrm{cop}}(n)$ has vertex set $[n]$ with $\{i, j\}$ an edge iff $\gcd(i, j) = 1$. Define $R_{\mathrm{cop}}(k)$ as the minimum n such that every 2-coloring of $E(G_{\mathrm{cop}}(n))$ contains a monochromatic K_k.

Since $E(G_{\mathrm{cop}}(n)) \subseteq E(K_n)$, any 2-coloring of K_n restricts to a 2-coloring of $G_{\mathrm{cop}}(n)$; if the former avoids monochromatic K_k, so does the latter (any monochromatic K_k in $G_{\mathrm{cop}}(n)$ would use only coprime edges, hence appear in K_n too). Therefore $R_{\mathrm{cop}}(k) \geq R(k, k)$ [6].

The clique number satisfies $\omega(G_{\mathrm{cop}}(n)) = \pi(n) + 1$: the set $\{1\} \cup \{\text{primes} \leq n\}$ is a clique of this size. For the upper bound, note that in any pairwise coprime set $S \subseteq \{2, \ldots, n\}$, each element's prime factorization uses a disjoint set of primes (else two elements share a factor); since each element uses ≥ 1 prime and all primes used are $\leq n$, we have $|S| \leq \pi(n)$.

Theorem 3.2. $R_{\mathrm{cop}}(3) = 11$.

Proof. By exhaustive extension from $n = 8$.

Base. Among all 2^{21} edge 2-colorings of $G_{\mathrm{cop}}(8)$ (21 edges), exactly 36 avoid monochromatic triangles.

Extension. For $n = 9$: vertex 9 adds 6 coprime edges; extending 36 colorings through 2^6 options each yields 36 avoiding. For $n = 10$: vertex 10 adds 4 edges; extension yields 156 avoiding. For $n = 11$: vertex 11 is prime, adding 10 edges (coprime to all of $1, \ldots, 10$); extending 156 colorings through 2^{10} options each yields 0 avoiding.

n	Edges in $G_{\mathrm{cop}}(n)$	New edges	Avoiding colorings
8	21	—	36
9	27	6	36
10	31	4	156
11	41	10	0

□

The increase at $n = 10$ reflects that $10 = 2 \times 5$ has only 4 coprime neighbors, imposing weak constraints. The collapse at $n = 11$ is driven by 11 being prime: its 10 edges create too many triangles for any 2-coloring to avoid.

Remark 3.3. $R_{\text{cop}}(3)/R(3,3) = 11/6 \approx 1.83$ exceeds the inverse coprime density $\pi^2/6 \approx 1.64$. If $G_{\text{cop}}(n)$ behaved like a random graph of density $6/\pi^2$, one would predict $R_{\text{cop}}(3) \approx 10$. The excess arises from multiplicative correlations among missing edges (multiples of a prime p form correlated non-adjacencies) [4].

Conjecture 3.4. $R_{\text{cop}}(4) = 20$. Evidence: *An avoiding coloring at $n = 19$ was found by random search ($\sim$160,000 trials); no avoiding coloring was found at any $n \in \{20, \ldots, 49\}$ ($\sim 10^6$ random trials each). This is heuristic evidence only—the extension method is infeasible at this scale ($\sim 1.47 \times 10^6$ base colorings).*

Open Problem 3.5. Determine the growth rate of $R_{\text{cop}}(k)$. Is $R_{\text{cop}}(k) > R(k,k)$ for all $k \geq 3$?

4 Coprime pairs in primitive sets

A set $A \subseteq [n]$ is *primitive* if no element divides another [2]. By Dilworth's theorem [1], every primitive $A \subseteq [n]$ has $|A| \leq \lceil n/2 \rceil$. Let $M(A) = |\{(a,b) \in A^2 : a < b,\ \gcd(a,b) = 1\}|$ and $M^*(n) = \max\{M(A) : A \subseteq [n] \text{ primitive}\}$.

Lichtman [5] proved that primes maximize $f(A) = \sum_{a \in A} 1/(a \log a)$ over primitive A—the Erdős primitive set conjecture. We show primes do *not* maximize M.

The *top layer* $T(n) = \{\lfloor n/2 \rfloor + 1, \ldots, n\}$ is primitive with $|T(n)| = \lceil n/2 \rceil$ (Dilworth-maximum). Define the *shifted top layer* $S(n)$ by replacing each even $2k \in T(n)$ with k when k is odd and $k > n/3$.

Lemma 4.1. *$S(n)$ is primitive with $|S(n)| = \lceil n/2 \rceil$.*

Proof. Replacement is bijective (each $2k$ maps to a distinct $k \in (n/3, n/2] \setminus T(n)$), preserving size. For primitivity: retained elements of $T(n)$ exceed $n/2$, so no two divide each other. A new element $k \in (n/3, n/2]$ cannot divide a retained element $b > n/2$: the multiple $2k$ was removed, and $3k > n \geq b$. Two new elements $k_1 < k_2$ with $k_1 \mid k_2$ would require $k_2 \geq 2k_1 > 2n/3 > n/2 \geq k_2$, a contradiction. □

Proposition 4.2. *As $n \to \infty$, $M(T(n)) \sim \frac{6}{\pi^2}\binom{\lceil n/2 \rceil}{2}$ and $M(S(n)) \sim \frac{64}{9\pi^2}\binom{\lceil n/2 \rceil}{2}$.*

Proof. For $T(n)$: the proportion of coprime pairs among L consecutive integers converges to $\prod_p (1 - 1/p^2) = 6/\pi^2$ as $L \to \infty$ (classical; see, e.g., Hardy–Wright [3], Thm. 332). Since $T(n)$ consists of $\lceil n/2 \rceil$ consecutive integers, the claim follows.

For $S(n)$: the even fraction satisfies $f_E \to 1/3$ (we replace roughly 1/6 of the $\lceil n/2 \rceil$ elements, reducing even elements from 1/2 to 1/3). Two even elements share factor 2, contributing 0 coprime pairs. For pairs with at least one odd element, coprimality is governed by odd primes; since the elements of $S(n)$ are drawn from intervals of length $\Theta(n)$ in $[n]$, the coprime probability for such pairs converges to $\prod_{p \geq 3}(1 - 1/p^2) = 8/\pi^2$. Weighting by parity classes: $d(S(n)) \to \frac{8}{\pi^2}(1 - f_E^2) = \frac{64}{9\pi^2}$. □

Theorem 4.3. *$M^*(n) \geq \frac{64}{9\pi^2}\binom{\lceil n/2 \rceil}{2}(1 + o(1))$. For $n \geq 10$, $M(P(n)) < M^*(n)$ where $P(n)$ is the set of primes up to n; the latter claim is verified by exhaustive computation.*

Proof. The lower bound is realized by $S(n)$ (Lemma 4.1 and Proposition 4.2). Since $M(P(n)) = \binom{\pi(n)}{2} \sim n^2/(2\ln^2 n) = o(n^2)$ while $M(S(n)) = \Theta(n^2)$, primes are eventually suboptimal. Exhaustive computation confirms this for $n \geq 10$:

n	Optimal primitive set	$M^*(n)$	$M(P(n))$
8	$\{2, 3, 5, 7\}$	6	6
10	$\{4, 5, 6, 7, 9\}$	8	6
15	$\{4, 6, 7, 9, 10, 11, 13, 15\}$	21	15
18	$\{4, 6, 7, 9, 10, 11, 13, 15, 17\}$	29	21

□

Conjecture 4.4. $M^*(n) \sim \frac{64}{9\pi^2}\binom{\lceil n/2 \rceil}{2}$. *That is, $S(n)$ is asymptotically optimal among primitive sets for maximizing coprime pairs.*

Remark 4.5. Optimal sets use prime powers ($4 = 2^2$, $9 = 3^2$) and products ($6 = 2 \cdot 3$, $10 = 2 \cdot 5$) instead of small primes. Primes have coprime density $d = 1$ but size only $\pi(n)$; Dilworth-maximum sets have size $\lceil n/2 \rceil$ but lower density. Since $M \propto d \cdot |A|^2$, size dominates.

Open Problem 4.6. Characterize which functionals F on primitive sets are maximized by primes. Lichtman's $\sum 1/(a \log a)$ favors primes; $M(A)$ does not.

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